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LAB 6(1)

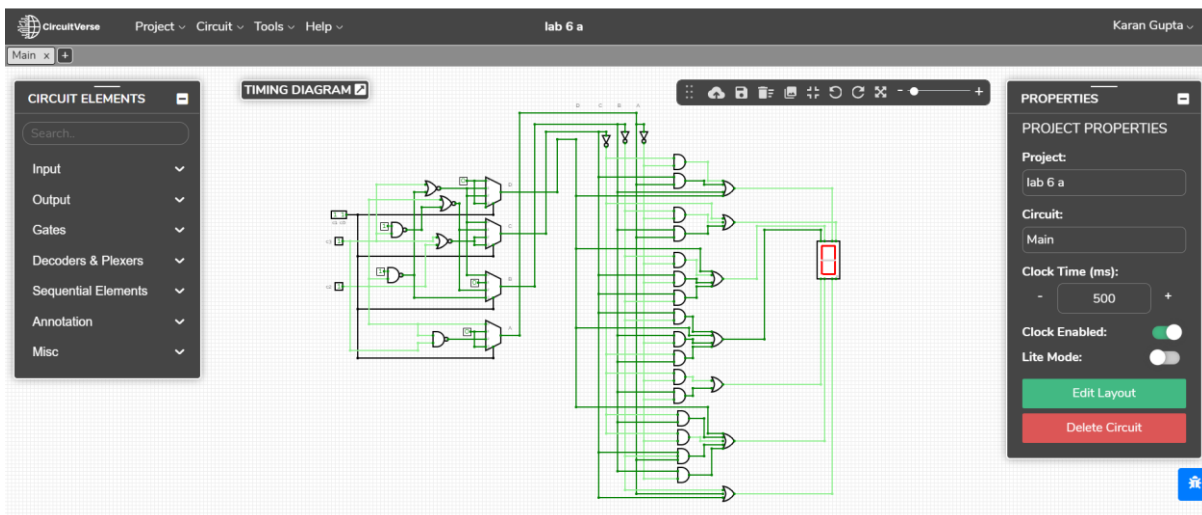
Aim: To design a decoder to recover the BCD representation(DCBA) of a decimal digit from C3 C2 C1 C0 representation(encoded representation)

Components Used:

1. Inputs
2. NAND gates
3. NOR gates
4. 4x1 mux
5. OR gates
6. AND gates
7. NOT gates
8. Seven Segment Display
9. Connecting wires

Link of circuitverse: <https://circuitverse.org/simulator/edit/lab-6-a-3e05113e-9b1f-4f0d-8647-7adaedb227d5>

Circuit Diagram:



Truth Table(Decoder for decoding code):

INPUT					OUTPUT				
DEC	C3	C2	C1	C0	DEC	D	C	B	A
0	0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	0	0
2	0	0	1	0	5	0	1	0	1
3	0	0	1	1	6	0	1	1	0
4	0	1	0	0	7	0	1	1	1
5	0	1	0	1	8	1	0	0	0
6	0	1	1	0	9	1	0	0	1
7	0	1	1	1	0	0	0	0	0
8	1	0	0	0	0	0	0	0	0
9	1	0	0	1	4	0	1	0	0
10	1	0	1	0	3	0	0	1	1

11	1	0	1	1	2	0	0	1	0
12	1	1	0	0	1	0	0	0	1
13	1	1	0	1	0	0	0	0	0
14	1	1	1	0	0	0	0	0	0
15	1	1	1	1	0	0	0	0	0

Truth Table(BCD to seven segment Decoder):

DEC	D	C	B	A	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	1	1	1	1
7	0	1	1	1	1	1	1	0	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	1	0	1	1

KMAPS(Decoder for decoding code):

D				
c3 c2 \ c1 c0	00	01	11	10
00	0	0	0	0
01	0	1	0	1
11	0	0	0	0
10	0	0	0	0

D				
c1 c0	00	01	11	10
0	0	c3'.c2	0	c3'.c2

C				
c3 c2 \ c1 c0	00	01	11	10
00	0	0	1	1
01	1	0	0	0
11	0	0	0	0
10	0	1	0	0

C				
c1 c0	00	01	11	10
c3'.c2	c3'.c2	c3.c2'	c3'.c2'	c3'.c2'

B				
c3 c2 \ c1 c0	00	01	11	10
00	0	0	1	0
01	1	0	0	0
11	0	0	0	0
10	0	0	1	1

B				
c1 c0	00	01	11	10
c3'.c2	c3'.c2	0	c2'	c3.c2'

A				
c3 c2 \ c1 c0	00	01	11	10
00	0	0	0	1
01	1	0	0	1

11	1	0	0	0
10	0	0	0	1

A				
c1 c0	00	01	11	10
	c2	0	0	c3'+c2'

KMAPS(Decoder for decoding code):

a

$a = \bar{c}\bar{a} + c\bar{a} + d + b$

b

$b = \bar{c} + \bar{b}\bar{a} + ba$

c

$c = \bar{b} + a + c$

d

$d = d + \bar{c}b + \bar{c}\bar{a} + c\bar{b}a + b\bar{a}$

e

$D_C \backslash BA$	00	01	11	10
00	1	0	0	1
01	0	0	0	1
11	x	x	x	x
10	1	0	x	x

$e = \bar{C}\bar{A} + B\bar{A}$

f

$D_C \backslash BA$	00	01	11	10
00	1	0	0	0
01	1	1	0	1
11	x	x	x	x
10	1	1	x	x

$f = D + \bar{B}\bar{A} + C\bar{A} + C\bar{B}$

g

$D_C \backslash BA$	00	01	11	10
00	0	0	1	1
01	1	1	0	1
11	x	x	x	x
10	1	1	x	x

$g = C\bar{B} + D + B\bar{A} + \bar{C}B$

Boolean Expressions: $a = C'A' + CA + D + B$

$$b = C' + B'A' + BA$$

$$c = B' + A + C$$

$$d = D + C'B + C'A' + CB'A + BA'$$

$$e = C'A' + BA'$$

$$f = D + B'A' + CA' + CB'$$

$$g = CB' + D + BA' + C'B$$

Observation: The decimal number was encoded using the following method:

$$C3C2C1C0 = 13 - N \text{ (in decimal) for } 0 \leq n \leq 4$$

$$C3C2C1C0 = N - 3 \text{ (in decimal) for } 5 \leq n \leq 9$$

Result: The decoder was successfully designed using 4x1 mux and the output was confirmed using seven segment display.

Application:

It is used in code conversion. For eg from analog to digital. It helps in converting decimal numbers to binary codes which are implemented in electronic circuits.

It is also used in data transmission

Decoders are used as address decoders in CPU memory location identification.

LAB 6(2)

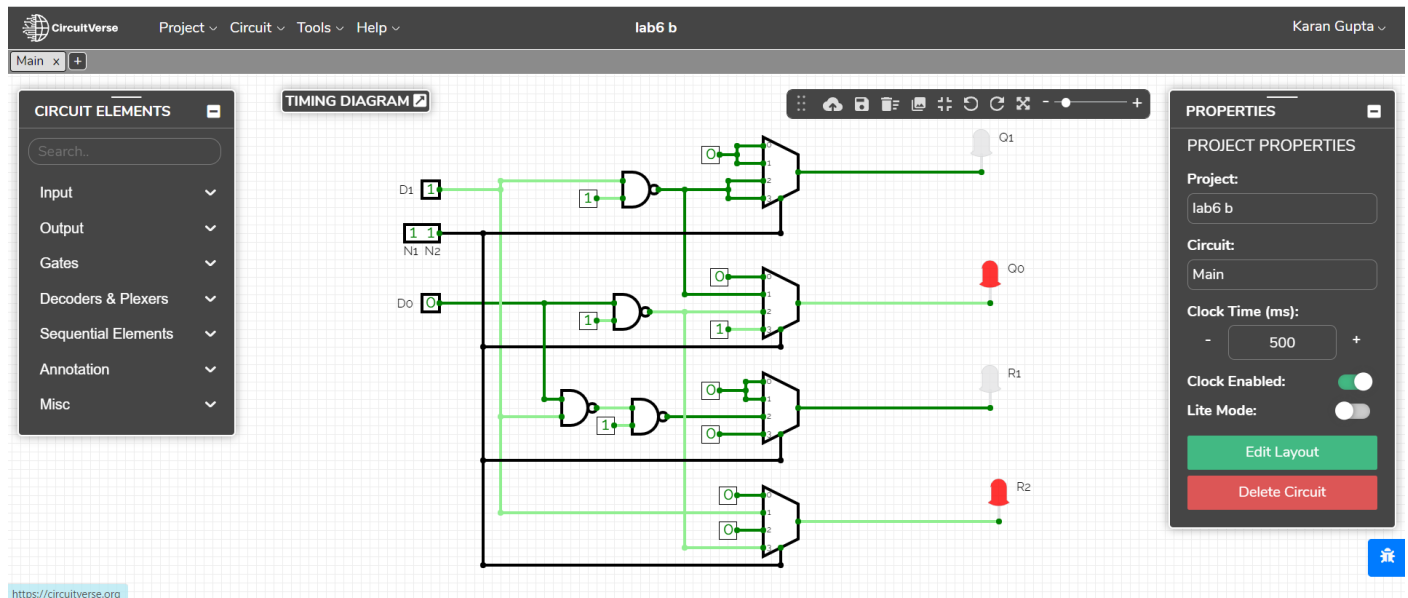
Aim: To design a combinational circuit for dividing a 2-bit number N1 N0 by another 2-bit number D1 D0 and (D1 D0 is not equal to 0) to generate a 2-bit quotient Q1 Q0 and a 2-bit remainder R1 R0.

Components Used:

1. Inputs
2. NAND gates
3. 4x1 MUX
4. LEDs

Link for Circuitverse: <https://circuitverse.org/simulator/edit/lab6-b-6780061e-03fb-4a25-9d8b-b49657e24184>

Circuit Diagram:



Truth Table:

N1	N0	D1	D0	Q1	Q0	R1	R0
0	0	0	0	X	X	X	X
0	0	0	1	0	0	0	0
0	0	1	0	0	0	0	0
0	0	1	1	0	0	0	0
0	1	0	0	X	X	X	X
0	1	0	1	0	1	0	0
0	1	1	0	0	0	0	1
0	1	1	1	0	0	0	1
1	0	0	0	X	X	X	X
1	0	0	1	1	0	0	0
1	0	1	0	0	1	0	0
1	0	1	1	0	0	1	0
1	1	0	0	X	X	X	X
1	1	0	1	1	1	0	0

1	1	1	0	0	1	0	1
1	1	1	1	0	1	0	0

KMaps:

Q_1

$N_1 N_0$ \ $D_1 D_0$	00	01	11	10
00	X	0	0	0
01	X	0	0	0
11	X	1	0	0
10	X	1	0	0

$N_1 N_0$	
00	0
01	0
11	$\overline{D_1}$
10	$\overline{D_1}$

Q_2

$N_1 N_0$ \ $D_1 D_0$	00	01	11	10
00	X	0	0	0
01	X	1	0	0
11	X	1	1	1
10	X	0	0	1

$N_1 N_0$	
00	0
01	$\overline{D_1}$
11	1
10	$\overline{D_0}$

R_1

$N_1 N_0$ \ $D_1 D_0$	00	01	11	10
00	X	0	0	0
01	X	0	0	0
11	X	0	0	0
10	X	0	1	0

$N_1 N_0$	
00	0
01	0
11	0
10	$D_1 D_0$

N ₁ N ₀	R ₀			
	00	01	11	10
00	X	0	0	0
01	X	0	1	1
11	X	0	0	1
10	X	0	0	0

N ₁ N ₀	
00	0
01	D ₁
11	$\overline{D_0}$
10	0

Boolean Expression:

$$Q_1 = N_1 \cdot N_0 \cdot D_1' + N_1 \cdot N_0' \cdot D_1'$$

$$Q_0 = N_1' \cdot N_0 \cdot D_1' + N_1 \cdot N_0 + N_1 \cdot N_0' \cdot D_0'$$

$$R_1 = N_1 \cdot N_0' \cdot D_1 \cdot D_0$$

$$R_0 = N_1' \cdot N_0 \cdot D_1 + N_1 \cdot N_0 \cdot D_0'$$

Observation: The truth table was verified using the divider circuit.

Result: The 2-bit division circuit was implemented successfully.

Application:

It is used in ALU for division operations